



Chapter Notes: Particulate Nature of Matter

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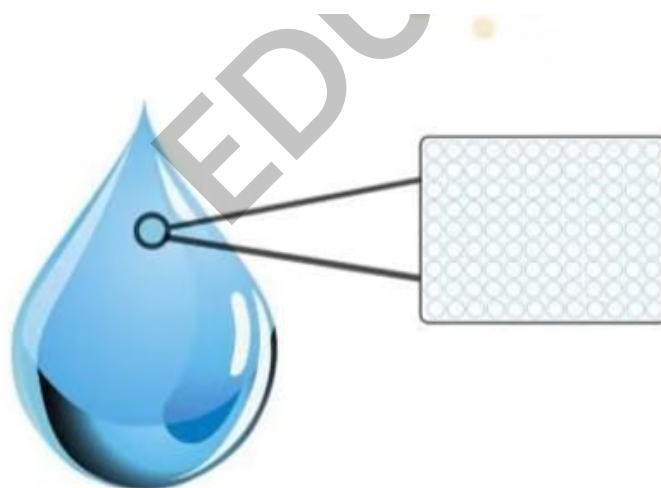
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Why can you stack a pile of sand, but not a puddle of water? This chapter will take you on a journey into the tiny building blocks that make up everything around us. You'll find out what **matter** really is, why solids, liquids and gases behave so differently, and just how small the basic particles of matter can be.

What Is Matter Composed of?

All matter is made up of tiny units called **constituent particles** or **basic building blocks**.



Matter is made up of tiny particles

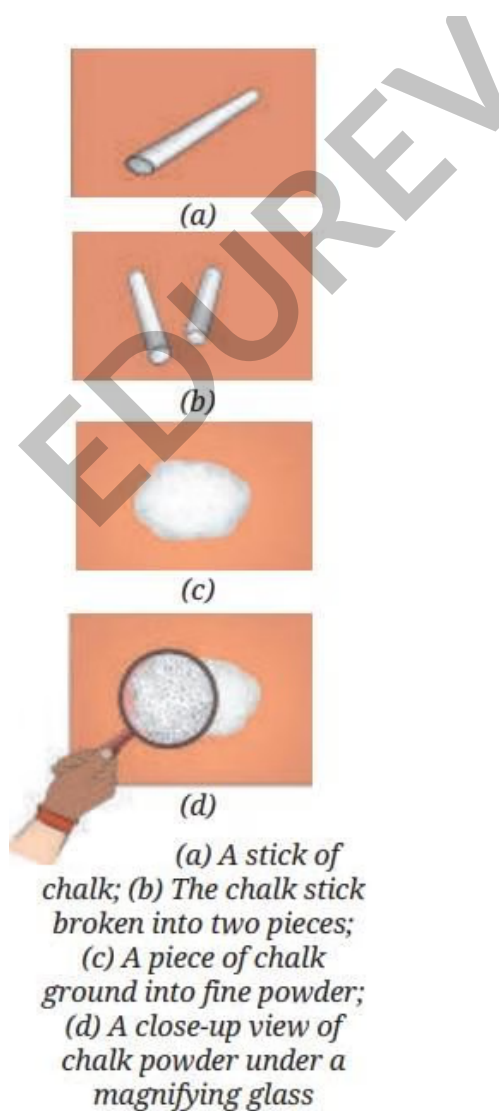
Example 1: Breaking Down Chalk

Breaking a piece of chalk into two, then into smaller and smaller pieces, and finally grinding it into fine powder shows that each piece still looks and behaves like chalk.

This shows that no matter how small the pieces become, each remains the same substance; breaking or grinding chalk is a **physical change**-no new substance is formed, only the size of pieces changes.

If you could continue breaking the chalk much further (well beyond ordinary sight), you would reach the **constituent particles** that cannot be broken down further by normal means.

Constituent particle: The basic unit that makes up a substance.



Example 2: Dissolving Sugar in Water

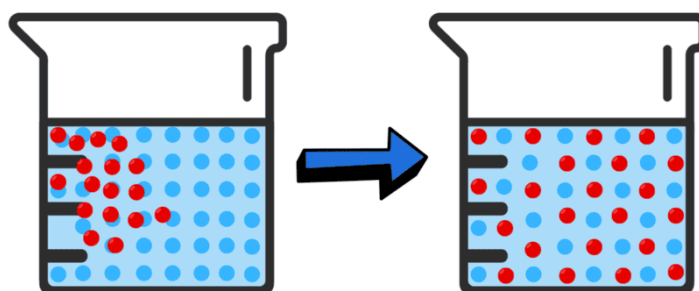
If sugar is put in water without stirring, the top layer may not taste sweet; after stirring the whole solution tastes sweet even though you cannot see sugar grains.

The sugar has not vanished; its particles are present but too small to be seen. The sugar has separated into its **constituent particles**, which spread among the water particles and occupy the available spaces (called **interparticle spaces**).



*Dissolving
sugar in water*

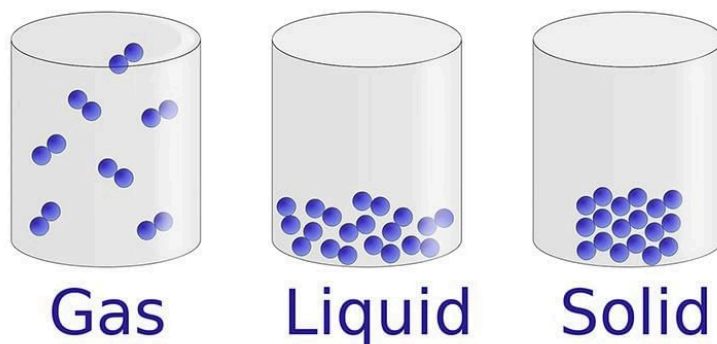
Particles of matter have spaces between them and so we can dissolve one form of matter over another



Conclusion: Both chalk and sugar can be broken down to pieces made of their basic particles, and these are so small they are invisible to the naked eye or even ordinary microscopes.

What Decides Different States of Matter?

The constituent particles of matter are held together by forces which are **attractive in nature**. These forces are called **interparticle attractions**.



Interparticle Attractions in Matter

The **strength of interparticle attractions** depends on how close the particles are and on the nature of the substance.

The **physical state** (solid, liquid or gas) of a substance depends mainly on the strength of these attractions and the distance between the particles.

Solid State

In **solids** particles are packed **tightly together**. The **interparticle attractions are very strong**, so particles cannot move freely - they only vibrate about fixed positions.



Some solid objects

Why are solids hard and keep their shape?

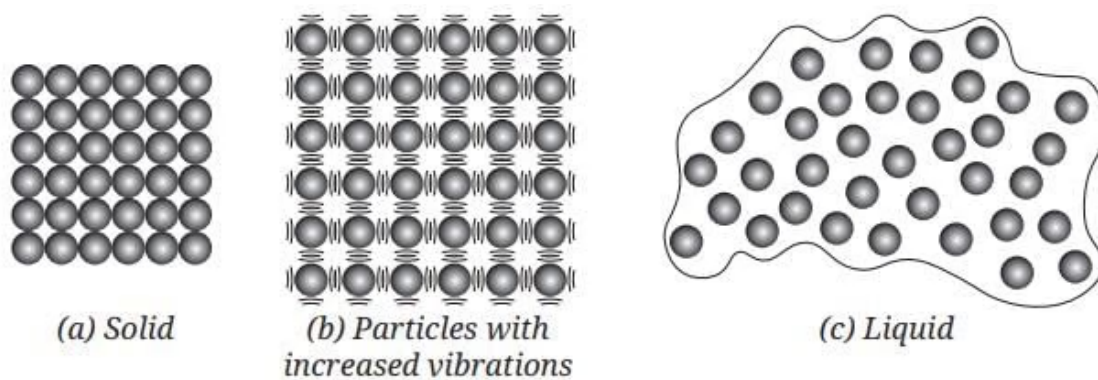
Solids have a **definite shape and a definite volume** because particles are tightly packed and the interparticle attractions are strong.

These strong attractions keep the particles in fixed positions, giving solids their hardness and fixed form.

When Solids Melt

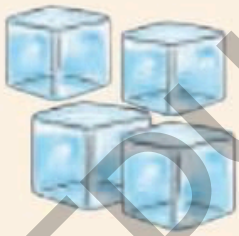
When a solid is heated its particles vibrate more vigorously.

At a certain temperature these vibrations become large enough for particles to leave their fixed positions and move past one another; the interparticle attractions weaken and the solid converts into a liquid.



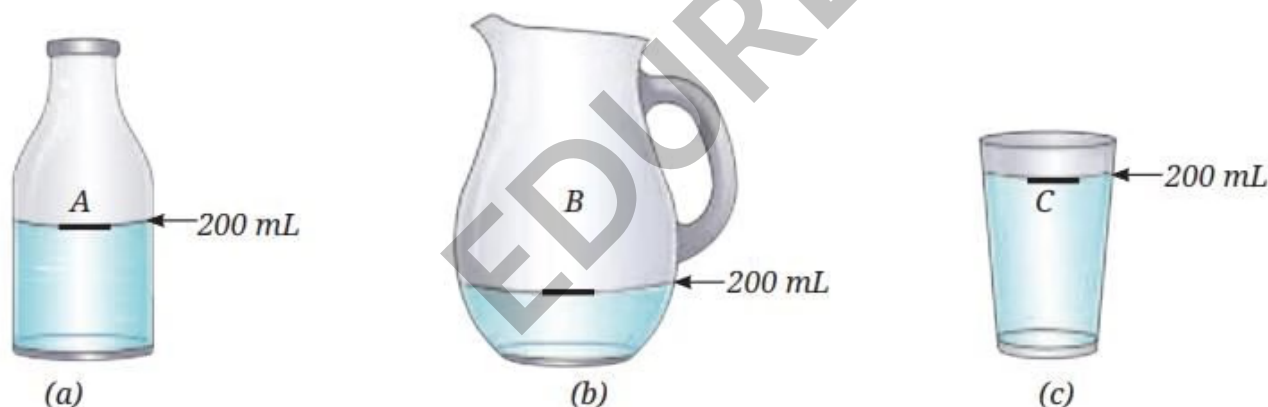
Magnified schematic pictures of melting of a solid

The temperature at which a solid changes into a liquid at atmospheric pressure is called its **melting point**.

S.No.	Material	Melting point
1.	Ice 	0 °C
2.	Urea 	133 °C
3.	Iron 	1538 °C

Liquid State

In **liquids** particles are **less tightly packed** than in solids. They remain close but can move past one another because the **interparticle attractions are weaker than in solids**.



Water placed in containers of different shapes

Why do liquids flow and take the shape of containers?

When water is poured from one container into another, it **takes the shape of the container**, while its volume stays the same.

Liquid particles can move past each other, so liquids have **no fixed shape** but do have a **fixed volume**.

Because particles are not fixed, you can move your finger through water; particles move but remain within the liquid's volume.



Heating Liquids

When a liquid is heated its particles gain energy and move faster.

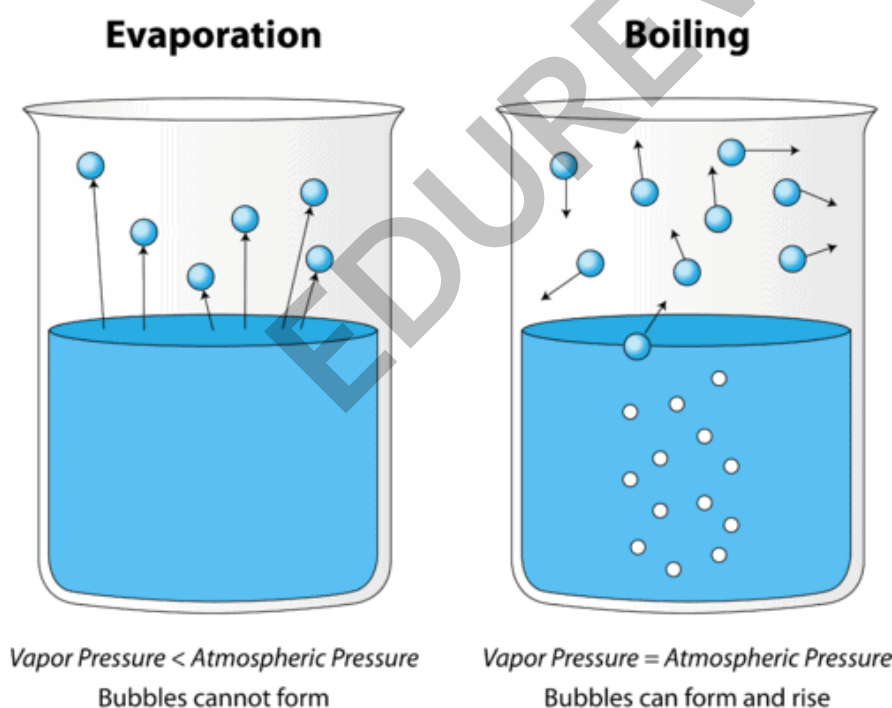
Eventually the liquid may **boil**, changing rapidly into vapor.

Key Definitions

Boiling point: The temperature at which a liquid boils and turns into vapour at atmospheric pressure; particle motion becomes very vigorous and they move apart.

Boiling: Rapid vapour formation that occurs at the boiling point and takes place throughout the liquid (visible as bubbles).

Evaporation: Slow vapour formation that can occur at temperatures below the boiling point; it occurs only at the liquid surface without bubble formation.



Particle behaviour during heating

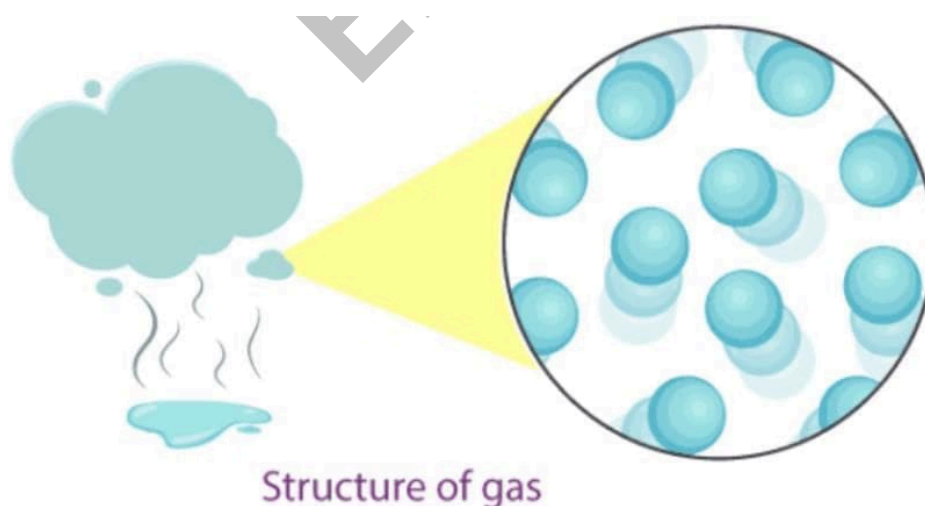
With heating, particles move more vigorously and separate from one another.

This reduces the effective interparticle attractions, allowing particles to escape the liquid and form vapour.

The overall transformation is the liquid converting to the gaseous state; boiling is the rapid, internal process and evaporation is the slower, surface process.

Gaseous State

In **gases** particles are **very far apart** and **interparticle attractions are almost zero**. Gas particles can **move freely in all directions**.

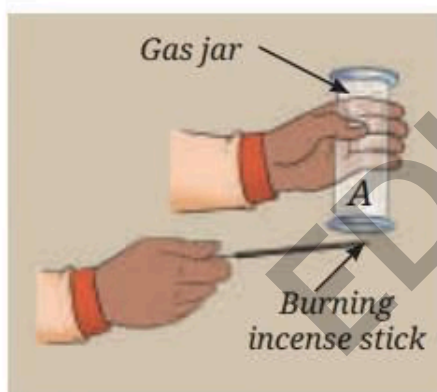


Why do gases have no fixed shape or volume?

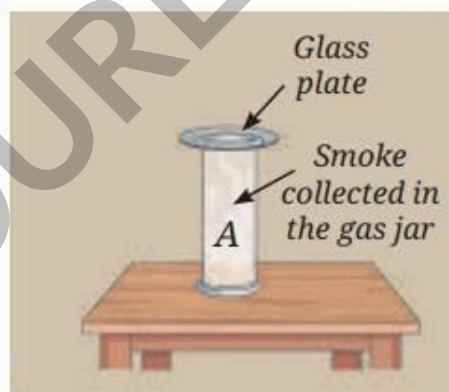
When smoke or any gas is allowed to move into another vessel it spreads and **fills the entire space** available to it.

Gas particles always spread out to completely fill any container; they have **no fixed shape** or **fixed volume**, and are highly compressible or expandable.

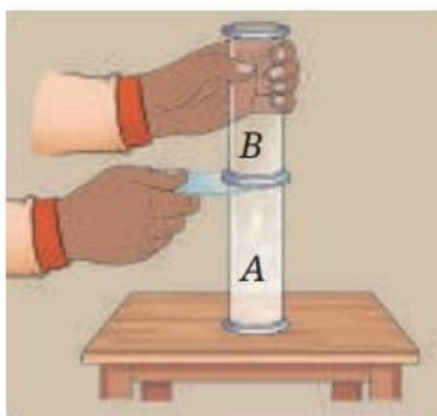
This is observed with smoke, iodine vapour and other gases: particles are in rapid, random motion and occupy all the space available.



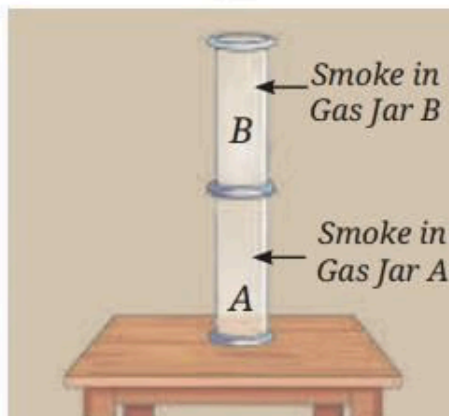
(a) Collecting smoke



(b) Covering the gas jar with a glass plate



(c) Placing the Gas Jar B and removing the glass plate



(d) Spreading the smoke

Smoke spreads freely inside the gas jars



Summary of the section:

Solids: Particles are packed tightly due to strong attractions - definite shape and volume.

Liquids: Particles can move past one another; weaker attractions than solids - take container's shape but have fixed volume.

Gases: Particles are far apart with negligible attractions - no definite shape or volume; fill the container and are compressible.

How Does the Interparticle Spacing Differ in the Three States of Matter?

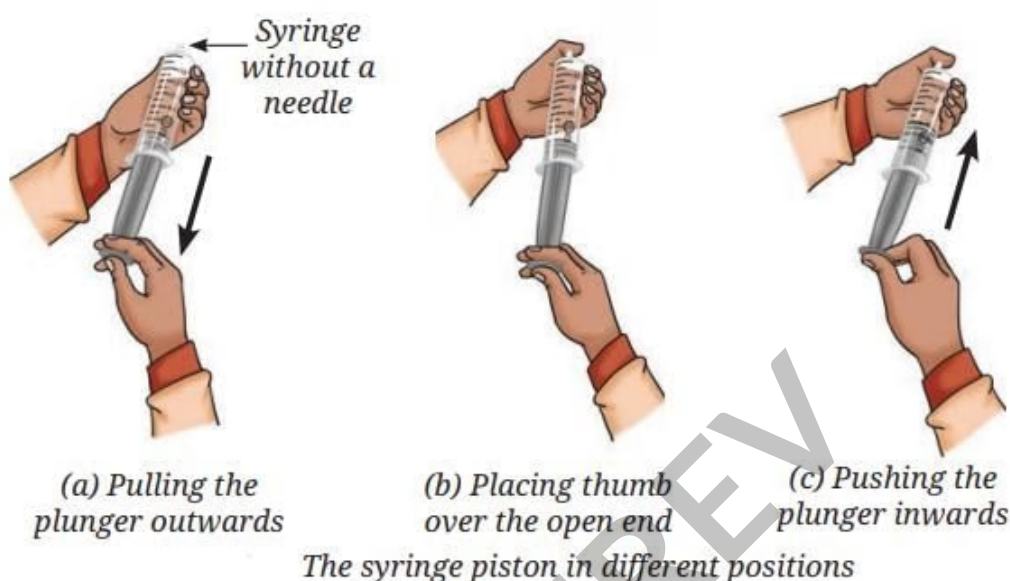
The **spaces between constituent particles** - called **interparticle spacing** - play a crucial role in determining whether a substance is solid, liquid or gas.

Smaller interparticle spacing: Stronger attractions, less movement - the substance tends to be a solid.

Larger interparticle spacing: Weaker attractions, more movement - the substance tends to be a liquid or gas.

Understanding Compression in Fluids (Syringe Experiments)

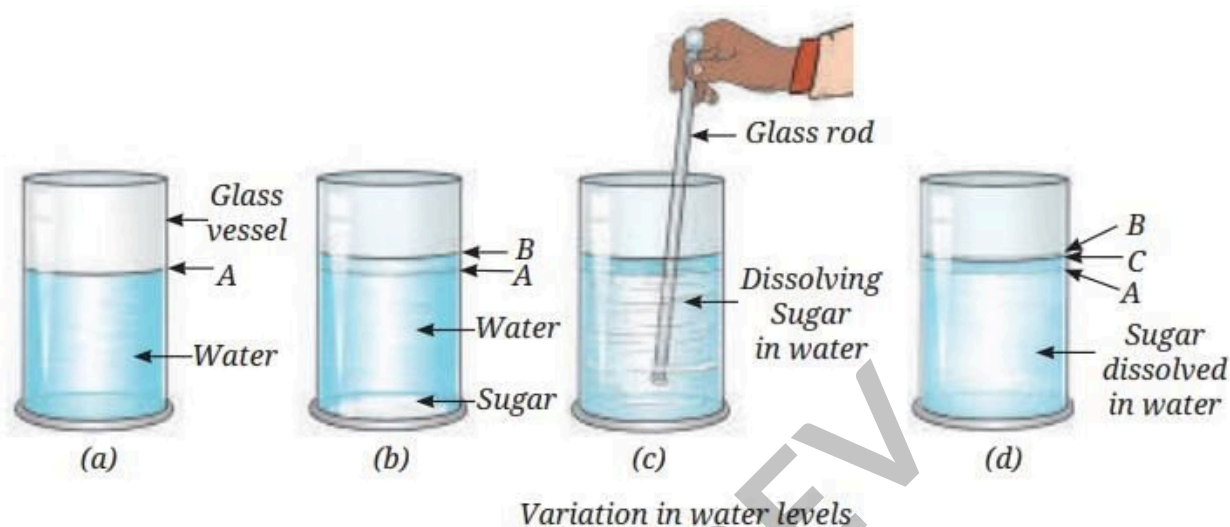
When you pull the plunger of a syringe, air fills the inside. If you place your thumb over the open end and push the plunger in, the **volume of air decreases**.



Gas particles have large gaps between them, so gases are **easily compressible**; compressing forces particles closer together.

If you repeat the experiment with water in the syringe (thumb on the end), the plunger cannot be pushed in much because liquids have **much smaller gaps** and are almost **incompressible**.

Dissolving Sugar - Evidencing Interparticle Spaces in Liquids



Add sugar to water: the water level rises as sugar is added, but after stirring the final liquid level is **less than the expected sum** of water plus sugar.

This shows that there are **empty spaces between water particles** (interparticle spaces); sugar particles occupy some of these spaces when dissolved.

If an insoluble solid such as sand is added, the water level stays high or rises because sand does not dissolve or fit into the interparticle spaces of water - sand particles simply add their own volume.

What About Solids?

In solids, strong attractions hold the particles **tightly together in a fixed arrangement**.

Interparticle spacing is very small - particles are closely packed and have very limited movement.

Although closely packed, solids still have a tiny amount of empty space between particles, but it is much less than in liquids or gases and contains nothing (not even air).

Scientific Meaning of "Particle"

In science, "particle" refers to the constituent unit that builds matter - atoms or molecules - which are far smaller than dust or sand grains.

In other contexts (for example, air pollution), "particle" may mean small dust bits, but even those are made of atoms and molecules.

Summary of this section

Gases: Easily compressed - lots of space between particles.

Liquids: Nearly incompressible - small interparticle gaps; dissolved substances can fit into these spaces.

Solids: Fixed, small gaps - particles packed tightly; hardly any compressibility.

MULTIPLE CHOICE QUESTION

Try yourself: What happens to gas particles when they are compressed?

- A They move farther apart.
- B They stay in the same place.
- C They are forced closer together.
- D They dissolve in liquid.

[View Solution](#)

How Particles Move in Different States of Matter

Particles are always in motion. The **movement of particles** is a key reason for differences in the behaviour of solids, liquids and gases.

Diffusion in Liquids (Potassium Permanganate Experiment)

Drop a crystal or a little potassium permanganate in water. At first you see coloured streaks spreading out; soon the whole glass of water becomes evenly coloured.

The reason is that water particles are constantly moving; they pull potassium permanganate particles from the grain and both kinds of particles keep moving and mixing until the colour is uniform.

Particles of liquids are **always moving**, and this motion causes dissolution and diffusion.



How temperature affects movement

The colour spreads faster in hot water, slower at room temperature, and slowest in ice-cold water.

More heat energy makes particles move faster.

Movement in Gases (Incense Stick Experiment)

Burn an incense stick in one corner of a room. Soon the fragrance spreads throughout the room, even to people far away from the stick.

The reason is that air particles are always moving rapidly in all directions, colliding with and carrying fragrance particles.

Particles of gases are in **constant, fast motion**, filling all available space and causing diffusion of gases and smells.

Why Some Substances Do Not Dissolve

Some solids (for example, sand) do not dissolve in water because their particles are held together too strongly and are not pulled apart or mixed by the moving water particles.

Other Real-Life Examples of Movement of Gases

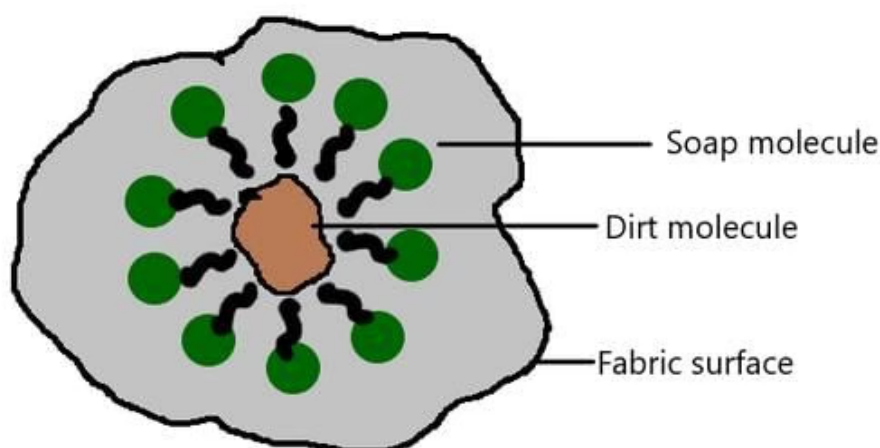
Cooking smells: Smell particles from cooking travel through the air to other rooms before you see the food.

Gas leak detection: A sharp smell from a broken container spreads rapidly, warning people even at a distance.

Smoke or fog: Smoke spreads and fills the space, showing how gas particles and suspended particles move.

Air fresheners: Plug-in or spray fresheners make a room smell pleasant quickly because their scent molecules travel via moving air particles.

How Soap Cleans Oily Stains (Particulate Nature in Action)



Soap molecules have two ends: one end is attracted to oil (hydrophobic) and the other end mixes with water (hydrophilic).

When soap is added to stained clothes, soap molecules surround (emulsify) oil droplets so that water can wash them away. This happens because soap particles interact with oil and water particles at the microscopic level.

This microscopic interaction and movement explain why washing with soap and water removes dirt and oil from clothes and skin.

Thermal Energy and Particle Movement

The thermal (heat) energy of particles determines how much they move and how far apart they are:

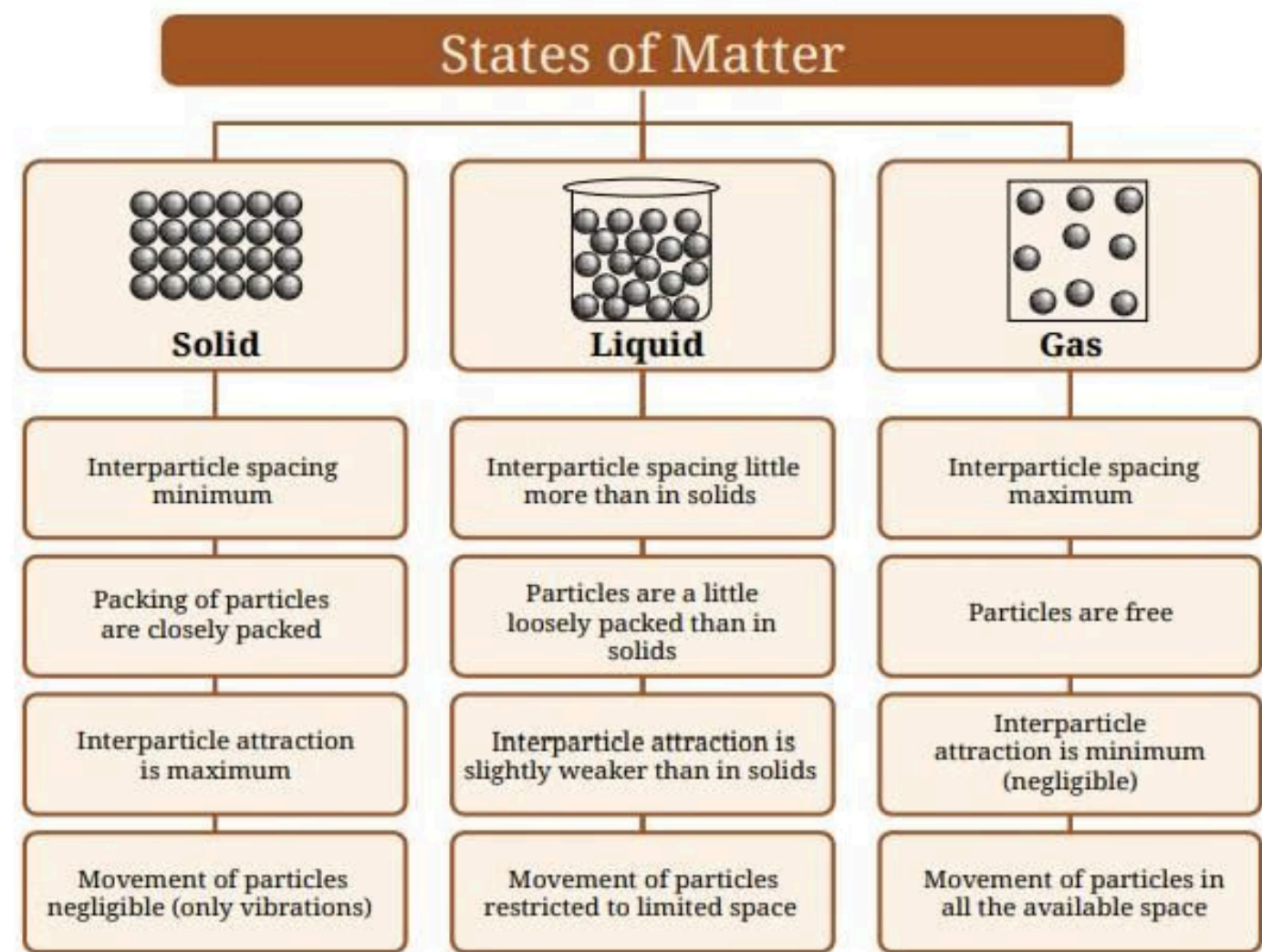
Solids: Lowest thermal energy; particles vibrate about fixed positions.

Liquids: More energy; particles move around each other but remain close.

Gases: Highest energy; particles move freely in all directions and are far apart.

Summary Table

Particle nature of the three states of matter -



Key Terms to Remember

Matter: Anything that has mass and occupies space.

Constituent particles: The tiny, basic building blocks that make up matter; they are extremely small and cannot be seen with the naked eye.

Interparticle forces of attraction: The attractive forces between particles. In gases these attractions are very weak, allowing free motion and spreading to fill containers.

Interparticle spacing: The distance between particles; minimum in solids, greater in liquids, maximum in gases.

Solid state: A state with strong interparticle attractions and minimum spacing; particles vibrate about fixed positions, giving fixed shape and volume.

Liquid state: A state with weaker attractions than solids, allowing limited particle movement and more spacing; liquids have fixed volume but no fixed shape.

Gaseous state: A state with negligible attractions, free particle movement and maximum spacing; gases have no fixed shape or volume.

Fixed shape: Property of matter that maintains its form (seen in solids).

Fixed volume: Property where matter occupies a definite amount of space (seen in solids and liquids).

Particle movement: Motion of particles - solids: vibration in place; liquids: movement past each other; gases: free and rapid motion - influenced by interparticle attractions and spacing.